

# AYUSH CHACHAN

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## EDUCATION

**Indian Institute of Technology, Delhi**

*Dual Degree (B. Tech and M. Tech) in Biochemical Engineering*

July 2016 – June 2022

CGPA - 8.90/10.00

## EXPERIENCE

**Fluxx Analytics** | *Founding Software Engineer*

June 2025 – Present

- **Event Ingestion & Data Pipeline**

- \* Architected a fault-tolerant event ingestion pipeline using **Kafka** consumers and **batch processing**, enabling SDKs to reliably **emit high-throughput** analytics events without strict schema constraints.
- \* Designed a custom **HTTP binary protocol** for event transmission, slashing **JSON**-based network overhead.
- \* Engineered a dual-storage data model utilizing **PostgreSQL** and **Redis** for rapid dynamic property lookups, alongside a column-oriented **BigQuery** pipeline to heavily optimize large-scale analytical queries.

- **Funnel Analytics Engine & DSL**

- \* Implemented a user-centric funnel analytics engine using **aggregated event arrays** instead of heavy self-joins, reducing evaluation complexity from  $O(n^2)$  to  $O(n)$  and cutting compute costs by **80%**.
- \* Designed a declarative **Funnel DSL** enabling configuration-driven execution, laying the groundwork for complex analytical features like occurrence constraints, hold properties, and multi-attribute splits.

- **Scalable User Segmentation**

- \* Designed a **hash-partitioned** user segmentation system leveraging **deterministic sharding** and **column-oriented storage** to ensure predictable performance across high-cardinality attributes.

**UPSC Civil Services Preparation**

May 2023 – May 2025

- Prepared for UPSC CSE with **Mathematics Optional**, involving sustained self-study and structured learning.

**Helloverify India Private Limited** | *Software Engineer*

July 2022 – Mar 2023

- Developed a **Django** verification platform utilizing a **Celery task scheduler** to trigger asynchronous **Selenium scrapers**, automating complex data extraction and generating consolidated **PDF reports**.
- Eliminated **15–30 minutes** of manual work per background check (scraping, tabulation, PDF assembly) across **20–50 daily checks**, replacing entirely manual workflows and cutting per-check turnaround by over **90%**.

## PROJECTS

**Mini Spring MVC Framework with Custom HTTP Server** | *Java, Networking, Concurrency*

- Built a **custom HTTP/1.1 server** from scratch, parsing raw **TCP byte streams** and utilizing a **thread-pool ExecutorService** to effectively enforce concurrent **thread limiting** and strict **backpressure**.
- Engineered a lightweight **IoC container** and **MVC Dispatcher** registry with custom annotations for automated component scanning, enabling a complete **Spring-like request pipeline** with **dependency injection**.

**Algorithms & Data Structures** | *Java, Graph Algorithms, Geometric DS*

Princeton COS226

- Modeled **percolation thresholds** on an  $n \times n$  grid via **Monte Carlo simulation**, applying **Weighted Quick Union-Find** with path compression for  $O(\log n)$  connectivity queries across thousands of randomized trials.
- Implemented **KD-Trees** to support efficient **range search** and **nearest-neighbor search** over 2D point sets.
- Implemented **content-aware image resizing** via seam carving using **dynamic programming** on a pixel energy **DAG**, computing and removing optimal vertical/horizontal seams to preserve salient features in iterative resizes.

## TECHNICAL SKILLS

**Languages & Frameworks:** Java, Python, JavaScript, C/C++, Bash, Spring Boot, Spring Cloud, Spring WebFlux.

**Databases & Messaging:** PostgreSQL, MySQL, MongoDB, Cassandra, BigQuery, Redis, Apache Kafka.

**Cloud, DevOps & Tools:** Docker, Kubernetes, AWS, Linux, CI/CD, Git, Maven, Selenium.

## ACHIEVEMENTS

- Awarded **IIT Delhi BOSS Award (2022)** for best experimental and design-based M. Tech project.
- **AIR 4334(GEN)(Top 0.4%)** in IIT-JEE Advanced 2016 among 1.2 M+ candidates in first attempt(Class 12th).